

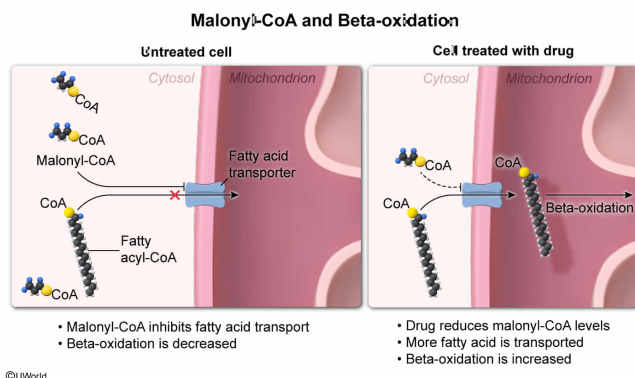
Malonyl-CoA is an intermediate in fatty acid synthesis and an inhibitor of long-chain fatty acid transport to the mitochondria. Given this information, which of the following experimental results would provide evidence that a certain drug causes cells to carry out increased beta oxidation?

- ☐ A. Adding malonyl-CoA to drug-treated cells causes an increase in the level of cytosolic acetyl-CoA compared to cells without malonyl-CoA treatment. [13%]
- ☐ B. Adding malonyl-CoA to drug-treated cells causes a decrease in the level of mitochondrial acetyl-CoA compared to cells without malonyl-CoA treatment. [22%]
- ✔ ☒ C. Treating cells with the drug results in a decreased concentration of cytosolic malonyl-CoA compared to untreated cells. [50%]
- ☐ D. Treating cells with the drug results in an increased concentration of cytosolic malonyl-CoA compared to untreated cells. [13%]

Correct

50% answered correctly

Explanation:



The effect of a drug on a metabolic pathway can be experimentally tested by measuring the levels of the molecules that participate in that pathway. As stated in the question, malonyl-CoA is an intermediate of **fatty acid synthesis** and an inhibitor of long-chain fatty acid transport to the mitochondria (a critical step in **fatty acid catabolism**). Consequently, changes in the level of malonyl-CoA in response to a drug could indicate changes in fatty acid metabolism.

The question asks about an experimental result that would provide evidence that a drug causes cells to carry out increased **beta oxidation** (ie, increased breakdown of fatty acids into acetyl-CoA in the mitochondria). Because malonyl-CoA inhibits long-chain fatty acid transport into the mitochondria, beta oxidation

is decreased in the presence of malonyl-CoA. If treatment with a drug led to **lower levels of cytosolic malonyl-CoA**, beta oxidation could **increase** as **more fatty acids enter the mitochondria**. Therefore, the observation that treatment with a drug results in a **decreased concentration of cytosolic**

malonyl-CoA would be one piece of evidence to suggest that the drug causes **increased beta oxidation**.

(Choices A and B) Adding malonyl-CoA to cells that had *already* been treated with the drug and *then* observing the effect on mitochondrial or cytosolic acetyl-CoA would only show the effect of adding malonyl-CoA, not the effect of the drug. To determine the effect of the drug, cells that were treated with the drug must be compared to cells that were not.

(Choice D) If the drug caused *increased* levels of cytosolic malonyl-CoA, it would most likely result in *decreased* beta oxidation because the additional malonyl-CoA would result in *less fatty acid transport* to the mitochondria.

Educational objective:

Malonyl-CoA helps regulate fatty acid metabolism by inhibiting long-chain fatty acid transport into the mitochondria. Processes that decrease malonyl-CoA levels are likely to increase fatty acid transport to the mitochondria and, as a result, increase beta oxidation.

Time spent:0 sec

Subject:
Biochemistry

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System:
Metabolic Reactions